

i-th Demonstration

What are the mean and standard deviation of a binomial experiment that occurs with probability of success 0.76 and is repeated 150 times?

A. 114, 27.35

B. 100.5, 5.23

C. 114, 5.23

D. The mean is 114, but there is not enough information given to determine the standard deviation.

Answer : D

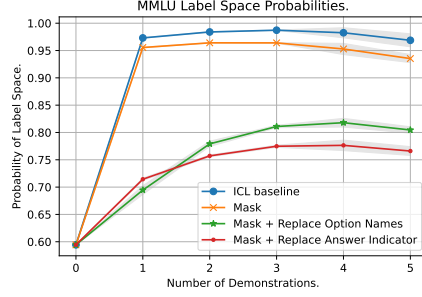


Figure 6: *Left: An example demonstration from MMLU’s high school statistics dataset.* Colors indicate the parts to be masked. *Right: Probability of MMLU’s label space.* We find that: (1) Masking question contents and answer contents of the demonstration does not influence directing the label space. (2) Both replacing the option names and the answer indicator significantly hurt the ability to constrain the label space. The gray shadow denotes the standard deviation across three runs.

further ablate option names and answer indicators by replacing them with semantically equivalent substitutes. More experimental details can be found in the Appendix B.

We use LLaMA-65B and plot the probability of choosing "A, B, C, D" as the predicted answer. The results are shown in the right of Figure 6. We find that masking the question and option contents, typically regarded as the most informative parts, does not influence the ability to direct the label space. Option names and answer indicators repeated several times in the demonstrations fulfill the job. Our conclusion is further solidified after ablating the option names and answer indicators individually. We see a huge decrease when replacing both option names and answer indicators, validating our hypothesis.

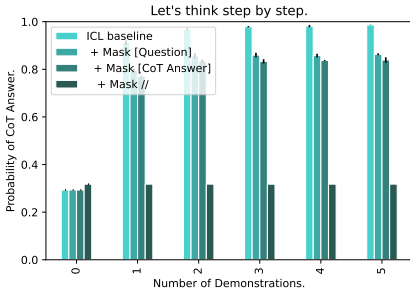


Figure 7: **Ability to follow patterns from demonstrations.**

Learning to Follow Patterns. Another distinctive feature of in-context learning is to follow the patterns of demonstrations. This is exemplified in techniques such as the Few-shot Chain-of-thought (CoT) prompting (Wei et al., 2022), frequently employed in reasoning tasks of LLMs like the GSM8K (Cobbe et al., 2021).

Here, we illustrate how the reinforced features in Section 3.1 affect the pattern following of ICL, by showing how LLMs follow the chain-of-thought demonstrations in the GSM8K high school math dataset. Each demonstration in the dataset follows the form “Question: [Question] // Let’s think step by step. // [CoT Answer]”. We demonstrate how models learn to say the CoT pattern, i.e., “Let’s think step by step.”. We further discuss the connection between surface patterns and [CoT Answer] in Appendix E.1.

Based on the findings in previous sections, we hypothesize that the common parts in the demonstrations teach the LLM to generate the CoT pattern. More specifically, ‘Question:’ builds a distant reinforcement, and the new liner ‘//’ builds a successive reinforcement with the CoT pattern. We mask out each part in each demonstration *progressively* with random tokens to ablate the influences.

The probability of the CoT pattern is shown in Figure 7. After masking out “//”, the probability gains obtained from demonstrations almost diminish, verifying our hypothesis of successive reinforcement. Another interesting finding is masking [Question] reduces the probability of generating the CoT pattern, indicating the [Question] part to some extent builds a connection with the CoT pattern. Since the questions in demonstrations are different but lie in the same group of grad math problems, there might be some common patterns among these questions.

Table 1: *Top: Effect of Non-Informative Connections (NC).* The accuracy of D increases with the cost of A, B and C. *Bottom: Effect of Reordered Answers.* With more reordered demonstrations, the outputs are more leaned toward D. The delta values on the superscript denote the improvement compared with zero-shot scenarios. ‘Avg. [A,B,C]’ denotes the average accuracy of samples whose golden answer is A, B or C. ‘D’ denotes the accuracy of samples whose golden answer is D. “†”: significantly different compared to its corresponding ICL baseline ($p < 0.05$).

<i>Non-informative Connections</i>						
# Demos	0	1	2	3	4	5
Avg. [A,B,C]	63.39	63.74 ^{+0.35}	64.56 ^{+1.17}	64.17 ^{+0.78}	64.44 ^{+1.05}	64.37 ^{+0.97}
Avg. [A,B,C] w/ NC	63.04	61.21 ^{-1.83†}	62.57 ^{-0.47†}	63.27 ^{+0.23†}	63.00 ^{-0.04†}	63.47 ^{+0.43}
D	52.28	59.65 ^{+7.37}	60.00 ^{+7.72}	59.30 ^{+7.02}	59.88 ^{+7.60}	59.53 ^{+7.25}
D w/ NC	49.47	63.63 ^{+14.15†}	64.09 ^{+14.62†}	63.51 ^{+14.04†}	62.57 ^{+13.10†}	61.64 ^{+12.16}
<i>Reordered Answers</i>						
Avg. [A,B,C]	63.39	63.74 ^{+0.35}	64.56 ^{+1.17}	64.17 ^{+0.78}	64.44 ^{+1.05}	64.37 ^{+0.97}
Avg. [A,B,C] w/ 50% RA	63.39	64.56 ^{+1.17}	64.44 ^{+1.05}	64.05 ^{+0.66}	63.94 ^{+0.55}	63.86 ^{+0.47}
Avg. [A,B,C] w/ 75% RA	63.39	64.52 ^{+1.13}	62.65 ^{-0.74†}	62.53 ^{-0.86†}	61.95 ^{-1.44†}	62.34 ^{-1.05†}
Avg. [A,B,C] w/ 100% RA	63.39	64.80 ^{+1.40†}	62.03 ^{-1.36†}	61.17 ^{-2.22†}	59.10 ^{-4.29†}	58.05 ^{-5.34†}
D	52.28	59.65 ^{+7.37}	60.00 ^{+7.72}	59.30 ^{+7.02}	59.88 ^{+7.60}	59.53 ^{+7.25}
D w/ 50% RA	52.28	59.30 ^{+7.02}	60.82 ^{+8.54}	61.40 ^{+9.12}	61.75 ^{+9.47}	61.64 ^{+9.36}
D w/ 75% RA	52.28	59.06 ^{+6.78}	62.81 ^{+10.53}	65.50 ^{+13.22†}	67.02 ^{+14.74†}	66.90 ^{+14.62†}
D w/ 100% RA	52.28	58.71 ^{+6.43}	66.20 ^{+13.92†}	71.35 ^{+19.06†}	75.67 ^{+23.39†}	77.19 ^{+24.91†}

4.2 DETRIMENTAL EFFECTS

Token reinforcement is not always helpful. In this section, we explore the detrimental consequences that arise from it. As shown in Section 3.1, distant and successive reinforcement are activated with even two random tokens. This could potentially lead to spurious patterns across demonstrations, which might be completely unforeseen by end users. We illustrate this point using two experiments where we manually construct spurious patterns in the MMLU dataset.

Non-informative Connections. Our first approach is adding connections between a phrase and a certain choice. We append a reasonable but non-informative phrase such as ‘*Please kindly provide your answer*’ or ‘*Looking forward to your choice*.’ right before the template ‘*Answer:*’ each time the question’s answer is ‘D’. In testing, we also append the same phrase and check whether the outputs are navigated toward ‘D’. By doing so, we construct a distant reinforcement loop from the non-informative phrase to the answer ‘D’. We ensure the test set is balanced with equal numbers of questions having golden answers “A,B,C,D”, and we report the accuracies at the top of Table 1.

We first see a gap even without demonstrations, where adding non-informative phrases lowers the accuracy of choices D. We further discuss the selection bias (Zheng et al., 2023) of different choices in the Appendix. Then, we see that the non-informative connection overcomes the selection bias and significantly elevates the accuracy choice D with a noticeable gap, in the cost of accuracy of A, B, and C. These results show the potential risk of manually injecting spurious connections and directing in-context learning toward unintended outcomes.

Answer Indicator Connections. In our second experiment, we show that reinforcing the connection between ‘*Answer:*’ and a certain choice, e.g., D, navigates the outputs. To this end, we randomly replace r percent of demonstrated answers with D. Simultaneously, we exchange the option contents of the original golden answer and D, to keep the demonstrations valid. In this way, we gradually reinforce the connection between ‘*Answer:*’ and ‘D’, with successive reinforcement.

The results are presented at the bottom of Table 1. Our baseline is 25%, where the demonstrations are balanced. With the increased ratio of answer D in demonstrations, the accuracy of D is largely improved, from 0.52 to 0.78, while the accuracy of A, B, and C decreases from 0.63 to 0.58. Our findings corroborate with An et al. (2023), where they show how diversity affects the performance of in-context learning. Our findings demonstrate how unbalanced demonstrations bring an unfair advantage for certain outputs.

Discussion (1) Reinforcements can be the underlying reason for ICL, but also causes its vulnerability. (2) Our observations guide how to build demonstrations to maximize ICL effectiveness! The demonstrations should be both balanced for all possible output labels, and kept concise enough without introducing any unnecessary reinforcement.

5 RELATED WORK

Explaining In-Context Learning. A range of contributions has deepened our understanding of In-Context Learning (ICL). Chan et al. (2022) and Xie et al. (2022) explore the emergence of ICL from the perspective of training data and Bayesian inference, respectively. Implicit learning of ICL over demonstrations is further highlighted by Garg et al. (2023), Li et al. (2023), and Hahn & Goyal (2023) theoretically show that performance of ICL can be represented by a complexity that repetition structures can be represented by a small PCFG tree. The similarity between gradient descent learner and in-context learner is demonstrated by Akyürek et al. (2023); von Oswald et al. (2023), while Dai et al. (2023) explain language models as meta-optimizers and likens ICL to implicit finetuning. Our work differs from this line of work with a novel perspective via repetitions and reinforced features, and our findings could potentially explain the mechanism of how LLMs achieve implicit learning. For instance, the step by step reinforcement across demonstrations intuitively resembles the gradient descent process of ICL described in previous work. Olsson et al. (2022) introduce induction heads of copying patterns and provide evidence for their relationship with ICL. Differently, our work investigates the LLM as a whole (Anderson, 1972), studies sophisticated patterns, views ICL as a combination of reinforcements, and scrutinizes both the benefits and drawbacks of reinforcements.

Analyzing In-Context Learning. Several studies have analyzed ICL properties. Min et al. (2022); Yoo et al. (2022) identify and discuss key factors that influence ICL capability such as input-label mappings. Wei et al. (2023) proposes that learning input-label mapping is an emergent ability. Factors such as structural similarity, diversity, simplicity (An et al., 2023), and order or embedding distribution (Lu et al., 2022; Liu et al., 2022) are also investigated. Pan et al. (2023) partitions ICL ability into task recognition and task learning, observing different phenomena with varying model sizes. Lastly, Si et al. (2023) unveils the presence of inductive biases in ICL by designing underspecified demonstrations.

Our findings corroborate with multiple previous analyses of in-context learning. For example, the scaling for distant reinforcement echoes Wei et al. (2023); Pan et al. (2023)’s findings of different phenomena when varying model sizes. The importance of demonstration ordering and diversity in An et al. (2023); Lu et al. (2022) can be explained by avoiding spurious connections.

Repetitive Generation and Self-Reinforcement Effect. Repetition is a notorious issue in neural text generation, affecting tasks like open-ended and directed generation (Holtzman et al., 2019; Welleck et al., 2019; Lin et al., 2021; See et al., 2017; Liu & Lapata, 2019). Maximization-based decoding strategies lead to bland, consecutive repetitions at word, phrase, and sentence levels (Holtzman et al., 2019; Welleck et al., 2019; Li et al., 2016; Karpathy & Fei-Fei, 2015; Guan et al., 2021). Despite advancements in large-scale pre-training with Transformer architecture (Vaswani et al., 2017; Radford et al., 2019; Lewis et al., 2020), unexpected sentence-level repetitions persist (Radford et al., 2019; Brown et al., 2020; Fu et al., 2020).

The repetition issue is puzzling given the lack of repetitive sentences in the training data. A series of studies investigate the cause, with both from the theoretical perspective (Fu et al., 2020) and empirical findings (Holtzman et al., 2019). Recently, Xu et al. (2022) proposes the *self-reinforcement effect*, suggesting a repetitive loop when combined with maximization decoding. Our study extends the effect to large language models and token reinforcement, explains the reasons, and bridges the excellent ability of in-context learning to this notorious issue.

6 CONCLUSION

We have taken a novel feature-centric approach to understanding in-context learning, by exploring its relationship with repetitive generation. We have identified a key mechanism, the token reinforcement loop, where any two tokens can form a strong connection through multiple co-occurrences. We delve into the reasons and inner-working of token reinforcement, demonstrating it to be an inevitable result of maximizing likelihood. Based on our findings, we view in-context learning as a combination of token reinforcements with different level of strength instead of input-label mapping. Furthermore,